A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The hydrosphere: hydrological cycle & unsustainable exploitation - Set 1

Mark scheme

AQA spec 3.2.2.1-3.2.2.3; The physical environment; textbook pp. 114-121

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 13 / 480 = 0.0271 years **1** 0.0271 x 365 **1** = 10 days (accept 9.9-10) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (days) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1 340 000 / 425 = 3153 years (accept 3100-3200) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working.	1
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** It has a very short residence time / very rapid transfer rate (480×10^{15} dm^3 per year) **1** So it moves large volumes of water each year, delivering precipitation to land – small reservoirs with fast transfers respond quickly to change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** The ocean is so large that an added or removed volume is a tiny proportion of the total **1** The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
01.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) **1** But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **transpiration** → water lost as vapour from the leaves **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Deforestation:** less interception so less evaporation from leaves – less transpiration so less water returned to the atmosphere and less precipitation downwind – more infiltration and surface runoff, higher river flows / flooding **2 Urban development:** impermeable concrete / tarmac reduce infiltration and aquifer recharge – runoff is faster and greater, increasing river flooding downstream – drains carry water quickly to rivers **2 Agriculture:** soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff – irrigation increases evaporation – loss of soil biota (worms) reduces infiltration and water retention – drainage lowers water tables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **infiltration** → water soaking into the soil surface • **impermeable** → does not let water through **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	6

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation **1** More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods **1** Changed wind patterns move humid air to different areas **1** Sea level rise / salt incursion into coastal aquifers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity **1** Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (68 000 - 7 000) / 68 000 x 100 **1** = 89.7% (accept 90%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued **1** Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake **1** Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) **1** Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) **1** Loss of livelihoods – conflict between the countries sharing the rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **salinisation** → salt building up in the soil / soil becoming salty **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Upstream abstraction for irrigation reduces the flow available downstream **1** Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (44 - 8) / 50 = 0.72 m yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation **1** The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer • **infiltration** → water soaking into the soil surface • **percolation** → water moving downwards through soil or rock	2
04.3	**1 mark each** 1 mark for each of the 3 points below. **1 Porosity:** a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) **1 Permeability:** the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) **1** Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **Permeability** → how easily water passes through / lets water through • **Impermeable** → does not let water through	3
04.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Salt water incursion:** seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) **1 Subsidence:** emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) **1** Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds **1 Ecological impacts:** plants needing high water availability die – wetland animals lost – food chain effects **1 Reduced supplies:** abstraction may suddenly have to stop after years of hidden decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter **1** Current recharge is little or none so any abstraction reduces the stored volume **1** The large volume means shortages take a long time to appear, but eventually the supply runs out – ancient water is often saline, causing soil salinisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Two satellites whose orbits are affected by the force of gravity **1** The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass **1 Advantage:** the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	3
05.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Low-pressure irrigation pipes to reduce leaks **1** Growing crops that need less water than wheat / maize **1** Charging consumers by volume used (metering) **1** Inter-basin transfer of water 450 km from the Yangtze **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

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Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
06.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2 Habitat change:** flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) **up to 2 Wildlife barrier:** the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river **up to 2 Changed river regime downstream:** regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds **up to 2 Sedimentation:** sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion **up to 2 Microclimate:** water's high heat capacity reduces temperature fluctuations – higher wind speeds – more evaporation, humidity, cloud and precipitation downwind **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vegetation / dead organic matter under water decays anaerobically **1** Releasing methane, a powerful greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

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Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rain surplus water is stored, reducing peak flow / flooding downstream **1** In dry periods the dam releases water to maintain river levels / supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** $3.6 \times 10^{11} / 1.2 \times 10^{9}$ **1** = 300 years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** It must be porous, with spaces between the particles, so that it can store a large volume of water **1** It must be permeable, so that water can flow through it to a well or spring fast enough to be abstracted **1** An impermeable layer beneath it stops the water draining away, and the rock must be exposed at the surface somewhere, or overlain by permeable material, so that rainfall can recharge it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **permeable** → how easily water passes through / lets water through • **impermeable** → does not let water through	3

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Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction of 4.8 x 10^9 m^3 per year is four times the recharge of 1.2 x 10^9 m^3 per year, so water is being removed faster than it is replaced **1** The stored water is being mined rather than used sustainably, so the saturated zone shrinks and its upper surface, the water table, falls each year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer	2
07.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Wells and boreholes must be deepened and the water pumped from further down, so the energy cost of abstraction rises and shallow wells fail first **1** Rivers, springs and wetlands fed by the aquifer dry up, destroying habitats and the species that depend on them **1** Near the coast, sea water is drawn into the aquifer (saltwater intrusion), making the water too saline for drinking or irrigation **1** The rock compacts as the water that supported it is removed, causing land subsidence that damages buildings and drainage, and permanently reduces the storage capacity **1** Irrigated agriculture that depends on the aquifer fails, with loss of food production and livelihoods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
07.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Fossil water fell as rain thousands of years ago under a wetter climate – under the present climate recharge is negligible, so any abstraction depletes a finite store and the resource is effectively non-renewable **1** Changing to crops with a lower water demand, or to drip / trickle irrigation applied at night, which delivers water to the roots and loses far less to evaporation **1** Metering and pricing water, or setting abstraction licences and quotas, so that users have a reason to waste less **1** Recycling treated waste water for irrigation or industry, and harvesting rainwater, so less is taken from the aquifer **1** Lining canals and repairing leaking pipes so that less of the water abstracted is lost before it is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer • **Recycling** → reprocessing a material to make something new **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box